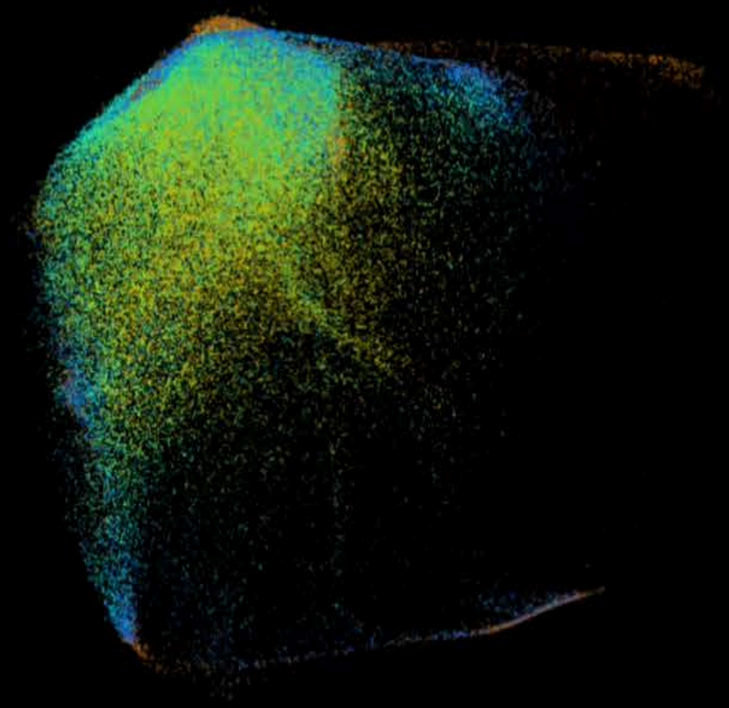
HEAR: TIME & DATA



DESN22848
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How Might We create an accessible experience that incorporates visual, auditory, and tactile senses through an objects distance, how can we allow affordability for individuals with sensory impairments?

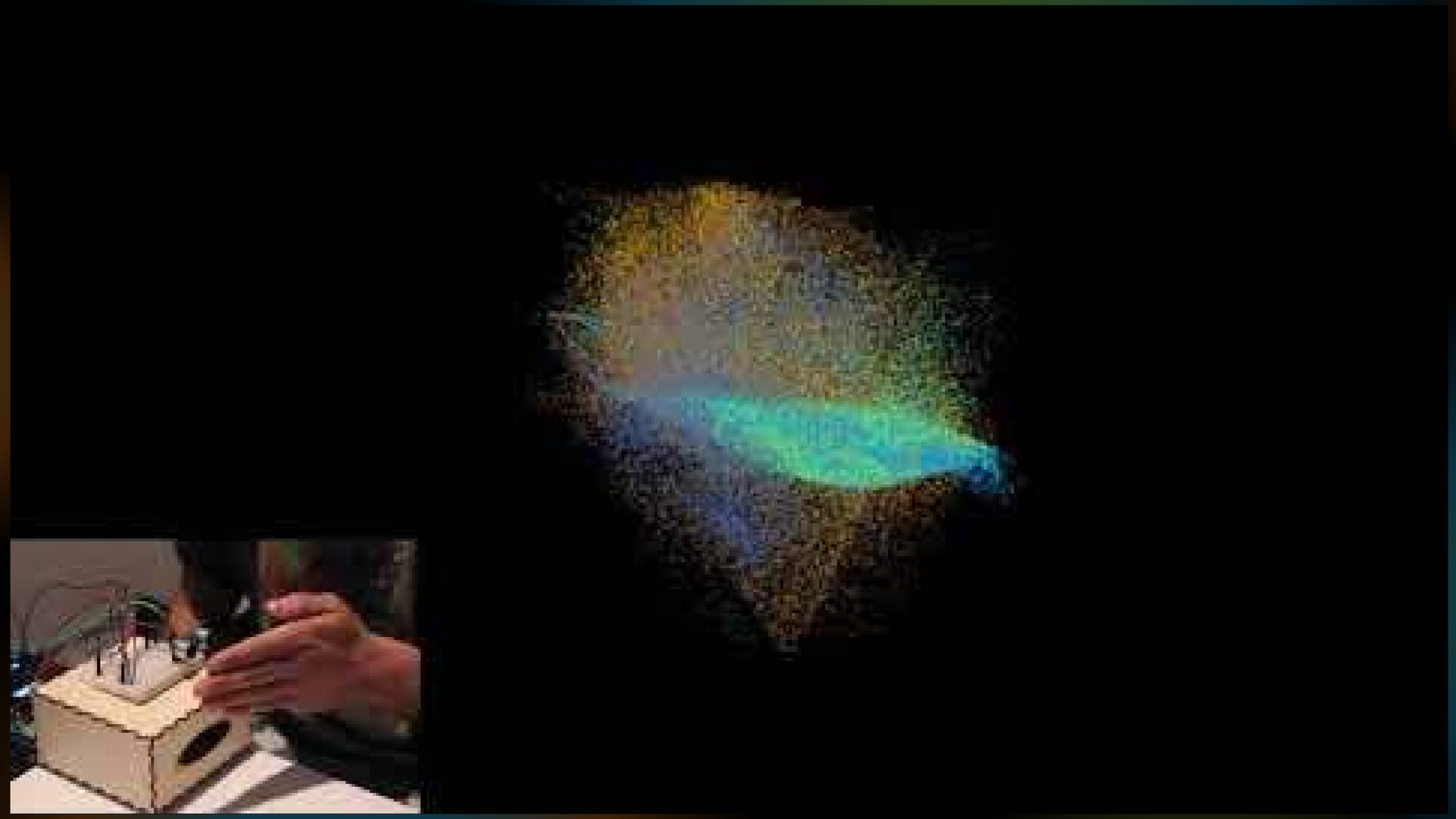
INDIVIDUAL DESIGN STATEMENT

In this project, **we explored how interactive media** can create a **more accessible listening experience** for individuals with **sensory impairments**. The concept originated from experimenting with sensor-driven interaction and generative media, with the goal of **transforming body movement** into a **multisensory** musical interface that connects users to sound through **motion**, **visuals**, and **touch**.

Using a **haptic sensor**, and an **ultrasonic sensor** connected to the **Arduino** and **TouchDesigner**, we developed a system that translates the distance of a user's hand into **dynamic audiovisual** and **tactile** responses. Through iterative testing, we mapped proximity data to control **sound levels**, **visual particle movement**, and **vibration** within our physical interface, **Haptic Zone**, allowing users to influence the environment through simple gestures rather than traditional controls.

By combining **sound**, **visual feedback**, and **haptic vibration**, the project explores an **alternative way of experiencing interactions** that **does not require technical training or precise motor control**. This multisensory interaction **lowers barriers** to participation and creates a more inclusive experience, allowing users to **hear**, **see**, and **feel sound** through the embodied interaction.

COLLABORATIVE DESIGN — FINAL



RESEARCH — CONCEPT 1

Through this project, I became interested in **how music and sound can be experienced beyond traditional listening**. Many musical instruments require fine motor control, technical training, and the ability to perceive sound clearly. This can **create barriers for beginners**, **children**, or **individuals with sensory impairments**.
In our class lectures, we discussed how **interaction design** increasingly **focuses on systems** that respond to the **human body**, **motion**, and **time** rather than static interfaces. The course describes **interaction with computational media** as something that **relies on human movement and physical interaction with digital systems**, creating experiences that **connect the physical and digital worlds**.

This idea directly influenced the direction of our project. Instead of interacting with sound through buttons or traditional instruments, **users interact through body movement**. The proximity of the user's hand becomes the input that shapes the audiovisual environment. Research in accessible music design also supports this approach. ***The Interaction Design Foundation*** explains that **inclusive design should consider the diverse sensory abilities of users and encourage designers to create multiple ways for people to perceive information,** rather than relying on a single sense such as hearing.

Similarly, research in inclusive music technology shows that translating sound into other sensory forms such as visual motion or vibration can **allow users to experience rhythm**, **intensity**, and e**motional qualities of music in alternative ways**. These systems **make music more accessible** for people with hearing impairments or limited musical training by allowing them to explore sound through movement and other sensory feedback.

RESEARCH — CONCEPT 2

Another concept that influenced this project is **embodied interaction**, which focuses on **designing digital systems that respond to human movement and physical presence**. In our program, we explored how **computational media** can respond to the **human body**, **motion**, and **time** rather than relying only on traditional screen-based interfaces. This approach allows interaction to extend beyond buttons or keyboards and instead use the body itself as part of the interface.

Movement-based interaction can also **support accessibility** by **reducing** the **need** for **precise technical controls**. When systems respond to gestures or **spatial movement**, users can interact more intuitively and explore the interaction through natural physical actions. Instead of requiring users to learn a complex interface, **embodied systems allow people** to discover how the system works through **experimentation** and **movement**.

The **W3C** explains that accessible systems should ensure information and interaction can be **perceived** and **operated** by users with different **abilities**. **Accessible design should not depend on a single form of input or interaction**, but **should allow alternative ways** for users to engage with digital systems.

Our project applies these ideas by using **proximity** and **gesture** as the **primary form of interaction**. The distance of the user's hand from the **ultrasonic sensor** controls the **volume** of the **ambient sound**, the **movement** of the **particle visuals**, and the **vibration** inside the physical box. Because the interaction relies on simple hand movement rather than precise controls, users can explore the experience in a more natural and **accessible way**. This demonstrates how embodied interaction can support inclusive design by allowing the body itself to become the interface.

RESEARCH — CONCEPT 3

One of the most influential ideas for this project came from Melissa Orta Martinez's TEDx talk *Haptic Technology in Education*. In her talk, Martinez challenges the way we typically think about sensory experience and learning. **She asks the audience to imagine what life would be like without the sense of touch**, pointing out that while **accessibility conversations often focus on vision or hearing**, the **tactile** sense is equally important in how humans understand and experience the world.

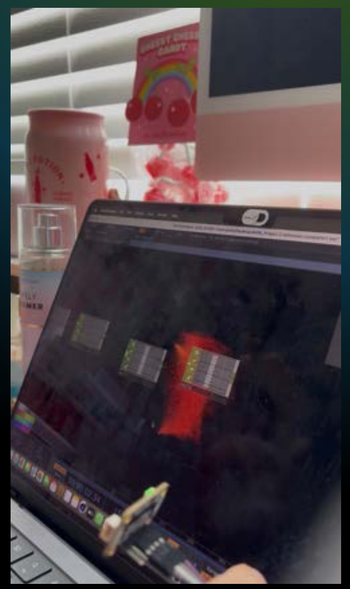
Martinez explains that **haptic technologies allow digital systems to communicate information through vibration and physical sensation**. For individuals with hearing impairments, **vibration can translate auditory information into tactile experiences**. Many deaf individuals describe feeling music through **low-frequency vibrations**, which allow them to perceive **rhythm**, **tempo**, and **emotional intensity** through their bodies rather than through hearing alone. This demonstrates that **sound is not only something we hear, but something we can physically experience through touch**.

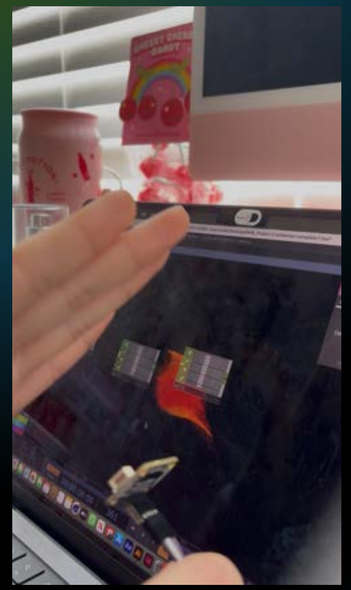
This concept strongly influenced the direction of our project. Rather than relying solely on audio output, we explored how haptic feedback could create an **additional** sensory channel within the interaction. Our system uses sensors connected to the Arduino to translate the distance of a user's hand into changes in sound levels and vibration inside our physical interface, **Haptic Zone**. As users move closer or farther away from the sensor, the environment responds through **both sound and tactile feedback**.

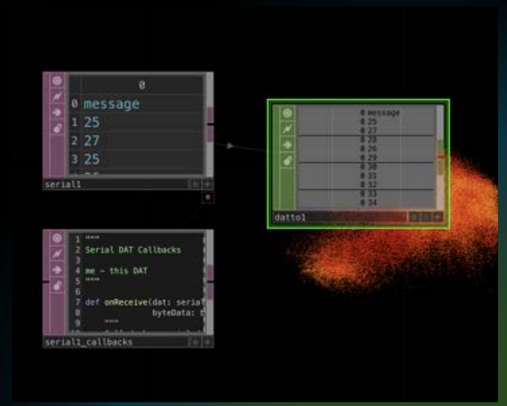
The use of haptic interaction also connects to ideas discussed in class about **embodied interaction**, where digital systems respond to **human movement** and **physical presence** rather than traditional interfaces like buttons or screens. By allowing users to interact with sound through **gesture** and **touch**, the system encourages a more **intuitive** and **physical** form of interaction. This approach demonstrates how haptic technologies can **support more inclusive interaction design**.

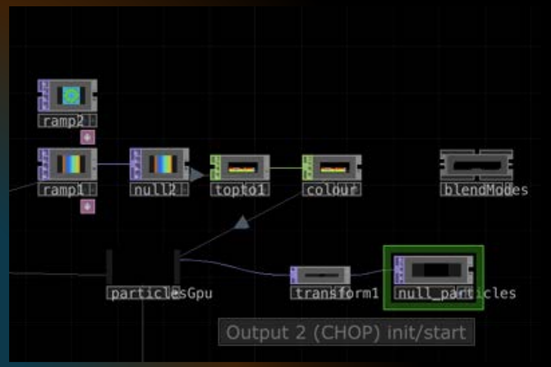
DESIGN IMAGE REFERENCES — PROCESS

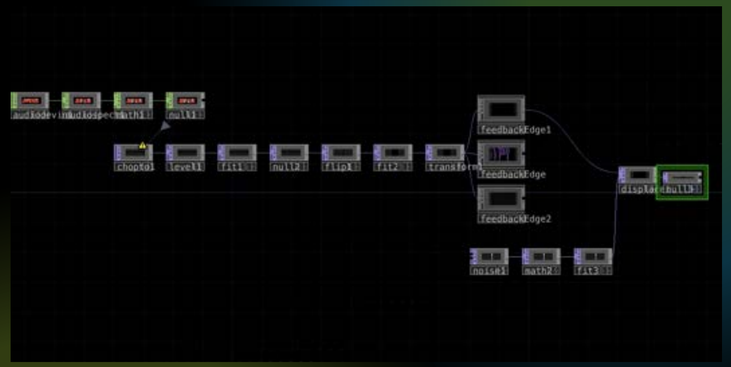
- Explored TouchDesigner networks and played around with the particle file Millie created
- Tested how audio levels could control visual parameters such as particle movement and intensity
- Experimented with different nodes and connections to create a visual environment that reacts to input
- Iterated on visual outputs so that the visuals would feel expressive and dynamic rather than static
- Began transitioning the system from audio-driven input toward sensor-based interaction with the ToF sensor





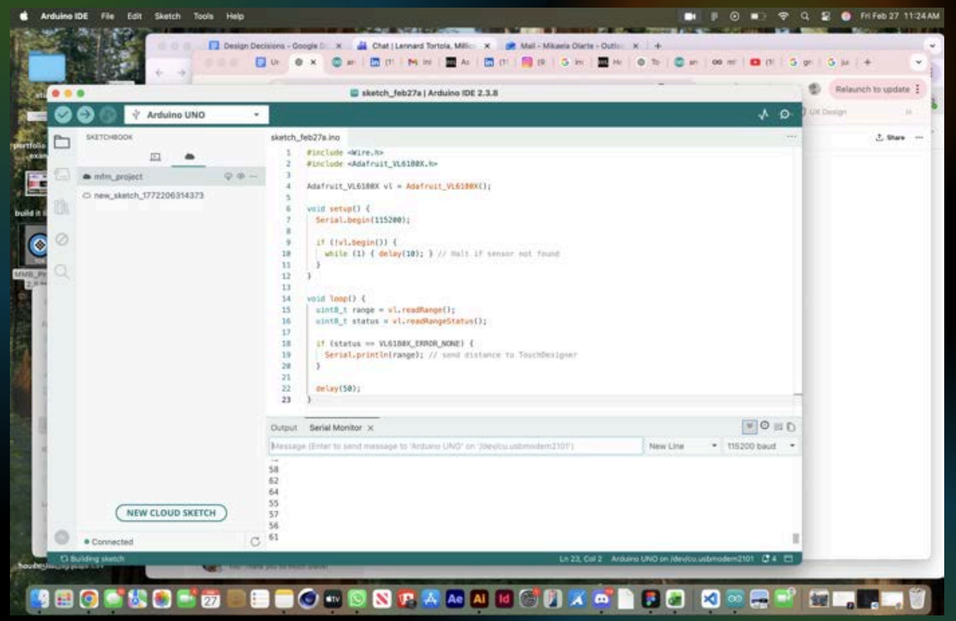


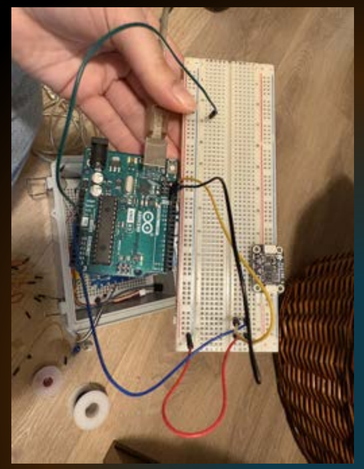


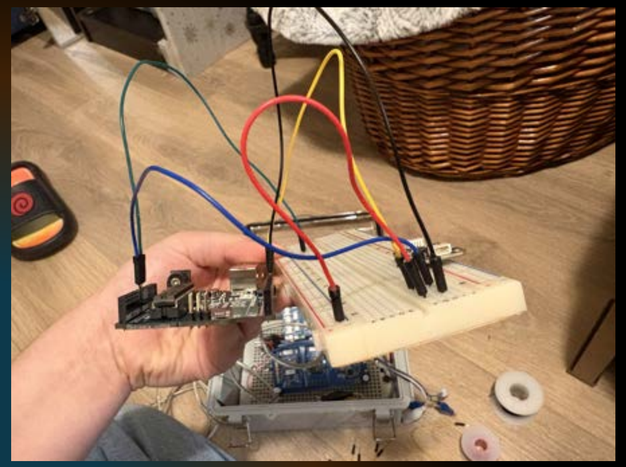


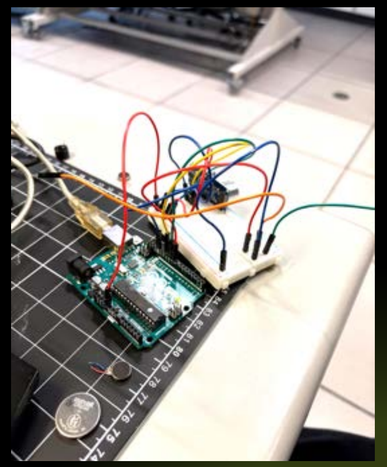
DESIGN IMAGE REFERENCES — PROCESS

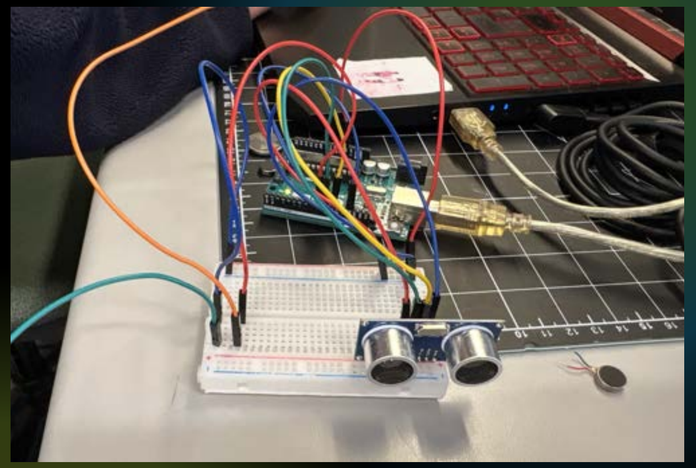
- Tested the original ToF distance sensor using Arduino code
- Experimented with how distance values changed as my hand moved closer or further away
- Initially wired the setup using alligator clips, which made the circuit harder to stabilize
- Later transitioned to using a breadboard with the ultrasonic sensor to create a more reliable setup
- Helped test how proximity data could eventually influence the interaction system











CHALLENGES

- Learning how to work with Arduino sensors and wiring, which was new to me
- Using alligator clips initially, which made the circuit unstable and harder to troubleshoot
- Understanding how sensor distance values translate into interaction data
- Learning how TouchDesigner nodes communicate and pass data through the network
- Translating Arduino sensor data into usable input for the visual system
- Debugging issues when visuals were still responding to audio instead of the proximity sensor
- Feeling less confident with the technical setup compared to the design side of the project

OVERCOMING CHALLENGES

- Used trial and error to explore TouchDesigner nodes and connections
- Collaborated with teammates to troubleshoot sensor data and system behaviour
- Tested components separately (sensor input vs. visuals) to better understand the system
- Experimented repeatedly with the sensor to understand how distance values behaved
- Moved from alligator clips to a breadboard to stabilize the circuit
- Tested Arduino code multiple times to observe how proximity values changed
- Continued iterating and testing until the proximity input began influencing the interaction correctly

REFLECTION

One of the biggest challenges in this project was understanding how to connect the physical hardware with the digital environment. Translating sensor data from the Arduino into TouchDesigner required learning how serial data works and how to convert it into usable parameters within a node-based system. Troubleshooting hardware connections and interpreting the sensor values also required patience and experimentation.

This project helped me better understand how physical computing can translate real-world motion into interactive digital experiences. I also gained confidence working with TouchDesigner and sensor-based systems, learning how data can be mapped to visual and sound responses. Conceptually, the project reinforced how embodied interaction can create intuitive and playful ways for users to engage with technology.

Moving forward, I would like to further refine how the sensor data influences the audiovisual experience, making the interaction smoother and more expressive. I would also explore expanding the accessibility aspect by experimenting with additional feedback such as vibration or alternative sound cues, helping the interaction become even more inclusive and engaging.

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WORKS CITED: CHATGPT

Description: Inquired how to connect the Time-of-Flight sensor and Arduino to TouchDesigner

"How do I connect the ToF sensor and Arduino to TouchDesigner so the distance data can control the visuals?" ChatGPT, OpenAI, 27 February 2026, https://chat.openai.com/.

Description: Inquired how to use sensor data to control interactive visuals

"I want to create an interactive light experience where the visuals grow when a hand comes near the sensor and reduce when the hand moves away." ChatGPT, OpenAI, 27 February 2026, https://chat.openai.com/.

Description: Inquired how to connect sensor interaction to sound and visual responses

"How do I map the sensor data in TouchDesigner so that particles and sound respond to the movement of my hand?" ChatGPT, OpenAI, 27 February 2026, https://chat.openai.com/.

Description: Inquired about using proximity interaction to create a musical or instrument-like interaction

"How can I make the interaction feel like you are playing an instrument using proximity and motion?" ChatGPT, OpenAI, 27 February 2026, https://chat.openai.com/.